

SHIVANI SWAMINATHAN

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EDUCATION

Georgia Institute of Technology

Atlanta, GA

Bachelor of Science / Master of Science in Aerospace Engineering

B.S. Fall 2027 | M.S. Fall 2028

GPA: 3.8/4.0 | Dean's List | Relevant Coursework: Aerodynamics, Structural Analysis, Fluids, Thermodynamics, Mechanics of Materials

RESEARCH

Ben T. Zinn Combustion Laboratory

Atlanta, GA

Undergraduate Research Assistant

May 2026 – Present

- Improved combustor-model fidelity by 18% across 8 steady and transient ANSYS Fluent cases by refining 19M-cell meshes and configuring reacting-flow physics for higher-confidence combustion predictions.
- Validated 3 combustor cases by analyzing temperature, pressure, velocity, and species results with ANSYS engineers, leading to 4 model and setup corrections and a more reliable simulation baseline.

GT Aerospace Systems Design Laboratory (ASDL)

Atlanta, GA

Undergraduate Research Assistant – Systems Modeling

August 2025 – Present

- Developed an executable SysML mobility subsystem linking torque, traction, speed, power, and climb constraints, then evaluated 30+ parametric cases to establish feasible performance limits.
- Built a traceable SysML model linking mission requirements, UAV architecture, operating behavior, and performance equations; evaluated 3 delivery scenarios against various range, battery, and payload requirements.

GT Aerospace Systems Design Laboratory (ASDL)

Atlanta, GA

Undergraduate Research Assistant – AI/ML

June 2026 – Present

- Developed a Python AI agent that expanded a lunar-base simulation to 12+ autonomous construction tasks, automating landing-pad sequencing, equipment allocation, and task dependencies for repeatable mission-plan evaluation.
- Integrated 4 lunar infrastructure models across 4 mission scenarios by defining dependencies among landing-pad construction, in-situ resource utilization (ISRU) mining, and power operations, enabling end-to-end surface construction simulation.

UGA Department of Physics and Astronomy

Athens, GA

Undergraduate Research Assistant

August 2024 – May 2025

- Analyzed 200+ stellar-flare light curves in Python and modeled activity decay across 3 spectral classes in MATLAB, reducing processing time by 36% and earning an undergraduate research award.

EXPERIENCE

GT HyTech Racing

Atlanta, GA

Aerodynamics Engineer

August 2025 – Present

- Validated carbon-fiber wing assemblies across 6+ ride-height and angle-of-attack cases in ANSYS Fluent, identifying up to 18% downforce and 11% drag variation to select competition wing ride height and angle of attack.
- Reduced analysis turnaround by 36% by automating ANSYS Fluent post-processing in Python, enabling same-day comparison of 8+ aerodynamic designs for faster design decisions.

Handshake AI

Remote

CAD Specialist – OpenAI-Sponsored AI Evaluation Project

March 2026 – Present

- Developed 25+ CAD benchmark tasks and weighted rubrics in SolidWorks, Siemens NX, and AutoCAD to evaluate AI-generated parts, assemblies, and drawings against measurable geometry, dimensioning, and deliverable criteria.
- Evaluated 40+ developer-created CAD benchmarks and identified 60+ dimensional, manufacturability, feature-quality, and deliverable issues, improving task clarity and scoring consistency before AI model testing.

PROJECTS

Wind Turbine Blade Fluid–Structure Interaction Analysis

July 2026

- Integrated CFD and FEA across 6 operating conditions, linking aerodynamic loads to blade stress and deformation while validating root radial force within 0.1% of hand calculations.

Unified Aerodynamic Analysis Tool

August 2026

- Developed a MATLAB aerodynamic solver combining Prandtl–Glauert-corrected thin-airfoil and shock-expansion theory, then searched 45,000+ configurations to optimize a double-wedge airfoil across subsonic and supersonic regimes.

CubeSat Frame & Attitude Control Design

January 2025 – March 2025

- Reduced CubeSat frame mass by 10% by iterating structural layout against 5g stiffness, safety-factor, and buckling constraints in ANSYS; developed a MATLAB/Simulink PD controller achieving 3-axis stabilization.

SKILLS

Simulation: ANSYS Fluent (CFD), ANSYS Mechanical (FEA), Simscape Multibody

3D Modeling: SolidWorks (CSWA), CATIA, Inventor, AutoCAD, Fusion 360, Siemens NX

Systems Engineering: MagicDraw/Cameo (SysML), ModelCenter, Simulink

Programming: MATLAB (Certified MATLAB Associate), Python, Git